

EDUCATION

- **Max Planck Institute for Informatics & Google (ELLIS Industry Track)** Saarbrücken, DE
PhD in Computer Science Sept 2024 - Current
- **Middle East Technical University** Ankara, TR
Bachelor of Science - Computer Engineering; GPA: 3.66/4.00, **top 5% of the class** Aug 2019 - June 2023
Activities: Founder of Management Consulting Club, Member of ACM Student Chapter, Former Member at Robotics Club

SELECTED PUBLICATIONS

- M. Farina*, V. Udandarao*, T. Nguyen*, **S. Kuzucu**†, M. Boether†, Andreas Hochlehnert†, Adhiraj Ghosh†, Marianna Nezhurina†, 29 more authors, “DataComp-VLM: Improved open datasets for Vision-Language Models”, under review, available at arXiv.
- **S. Kuzucu**, A. Tonioni, V. Lup, B. Schiele, F. Tombari, M. F. Naeem, “PARCEL: Pool-Anchored Resampling with Conditioned Elastic Queries for Efficient Vision-Language Understanding”, under review, available at arXiv.
- **S. Kuzucu**, M. F. Naeem, A. Kukleva, F. Tombari, B. Schiele, “Enhancing Self-Supervised Visual Representation Learning via Low-Rank Adapted LLMs”, under review.
- **S. Kuzucu***, K. Oksuz*, J. Sadeghi, P. K. Dokania, “On Calibration of Object Detectors: Pitfalls, Evaluation and Baselines”, selected for **Oral Presentation (top 2.3% of submissions)** at **European Conference on Computer Vision (ECCV) 2024**.
- K. Oksuz, **S. Kuzucu**, T. Joy, P. K. Dokania, “MoCaE: Mixture of Calibrated Experts Significantly Improves Object Detection”, at **Transactions on Machine Learning Research (TMLR)**, reviewed on Openreview.
- **S. Kuzucu**, J. Cheong, S. Kalkan, H. Gunes, “Uncertainty-based Fairness Measures”, at **ACM Journal of Artificial Intelligence Research (JAIR)**.

EXPERIENCE

- **Google Zurich** September 2025 - Present
Student Researcher **Supervisors: Dr. Ferjad Naeem & Dr. Federico Tombari**
 - **Research Topic** - Efficient and elastic inference with vision-language models
- **Max Planck Institute for Informatics** September 2024 - Present
Doctoral Researcher **Advisors: Prof. Dr. Bernt Schiele & Dr. Federico Tombari**
 - **Research Topic** - Efficient visual representation learning with pretrained LLM representations
 - **Contributions:** Lead the project on leveraging off-the-shelf LLM representations on improving vision-only tasks.
- **Five AI (Bosch UK)- Oxford Applied Research Center** September 2023 - June 2024
Intern Research Scientist **Supervisor: Dr. Puneet K. Dokania**
 - **Research Topic** - Calibration and Evaluation of Object Detectors
 - **Contributions:** Lead the project on proper performance and reliability benchmarking of object detectors. Showed that simple post-hoc calibrators can beat the *state-of-the-art* training-time calibration techniques by more than 90%. Developed and released an open-source library to allow broader community usage, available at GitHub.
- **AFAR Lab at University of Cambridge** February 2022 - August 2023
Undergraduate Student Researcher **Supervisor: Prof. Dr. Hatice Gunes**
 - **Research Topic** - Uncertainty-based Fairness Measures
 - **Contributions:** Lead the project and showed that the existing point-based fairness measures can cause potential pitfalls to miss the existing biases. Introduced a new notion for fairness gap based on uncertainty discrepancies across subgroups, and evidenced their utility through extensive experiments on three synthetic datasets and three real-life datasets.
- **METU Image Lab** October 2021 - September 2022
Undergraduate Student Researcher **Supervisor: Prof. Sinan Kalkan**
 - **Research Topic** - Uncertainty As A Measure to Mitigate Class Imbalance
 - **Contributions** Worked with various modern uncertainty quantification methods to overcome the class imbalance problem. Furthermore, curated a novel semantically-imbalanced dataset called “SVCI’20” and benchmarked the state-of-the-art imbalance mitigation techniques with it.

PROJECTS

- **Code Implementation for Uncertainty-Aware Learning Against Label Noise on Imbalanced Datasets (AAAI'22)** Provided the unofficial implementation for all of the methods described in the work as the official code was not released. Conducted as part of the graduate course CENG502: Advanced Deep Learning. Available at GitHub (June, 2023)
- **Code Implementation for Uncertainty Quantification in CNN Through the Bootstrap of Convex Neural Networks (AAAI'21)** Provided the unofficial implementation for all of the methods described in the work as the official code was not released. Available at GitHub (June, 2022)

HONORS, AWARDS & TEST SCORES

- Earned Turkish Presidency Extraordinary Young Talent Award for my studies and activities during both college and high school - September 2023
- Scored 328/340 in GRE (Verbal: 160/170 (85%), Quantitative: 168/170 (90%), Analytical Writing: 4.5/6 (79%)) - October 2022
- First place in Guided Research Symposium at METU Computer Engineering Department out of 30 different undergraduate research projects with my work in developing a novel one-pass uncertainty quantification method - June 2022
- Finalist (Top 10 out of 150 teams) at the Swarm Robotics Competition at Teknofest 2021- August 2021
- Scored 118/120 in TOEFL (R: 30, L: 29, S: 30, W: 29) - August 2021
- Earned METU Development Foundation's Academic Merit Scholarship - September 2019
- Ranked in top 0.01 % (270th) at Turkish university entrance exam amongst 2.5 million test takers - July 2019

SKILLS SUMMARY

- **Fluent Languages** Python, C++, JavaScript, C#, C, Bash, MySQL
- **Frameworks and Platforms** PyTorch, MMDetection & MMCV, ROS, Angular, Flask, Linux
- **Miscellaneous** Playing bass and electric guitar, playing Baduk (or Go-Weiqi), learning about Chinese language and culture (Mo Yan, Yu Hua and Cixin Liu are some of my favorite), reading-researching on gothic and far-eastern literatures